

SMART INDIA HACKATHON 2026



- Problem Statement ID – SIH26077
- ProblemStatement Title- AI-Driven
Hyper-Local Early
Warning System for Severe Weather
Nowcasting
- Theme- MedTech/BioTech/HealthTech
- PS Category- Software
- Team ID-
- Team Name (Registered on portal)- DEBUGGERS



❑ AGNI-CAST — *AI-Driven Hyper-Local Weather Forecast & Alert System*



PROPOSED SOLUTION :

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- ❑ Multi-source weather and terrain data from **INSAT-3DR, IMDAA, GSMaP and DEM**.
- ❑ **WV, TIR-1, TIR-2 and MIR** bands for moisture and cloud/thermal monitoring.
- ❑ Analysis of **temperature, humidity, CAPE, CIN, wind shear and rainfall**.
- ❑ **Multi-horizon AI nowcasting** using ConvLSTM + Spatial-Temporal Transformer.
- ❑ **0–100 hyper-local risk scoring** on a 1–3 km risk grid.
- ❑ Admin-approved multilingual alerts with **recommended action and safe-shelter guidance**.

WHAT MAKES AGNI-CAST DIFFERENT

- ❑ **Multi-Modal AI Fusion**
- ❑ **Explainable Hyper-Local Risk**
- ❑ **Predictive Storm & Flood Impact Mapping**
- ❑ **RAG + Human-Verified Decision Support**
- ❑ **Multilingual Actionable Alerts**

REQUIREMENTS

Data Requirements

INSAT-3DR satellite observations
IMDAA atmospheric data
GSMaP rainfall data
CartoDEM/SRTM elevation data

Technology Requirements

Python + PyTorch/TensorFlow
Spatio-Temporal AI model
PostgreSQL/PostGIS
FastAPI + Web dashboard

KEY FEATURESs

1. Multi-Source AI Weather Analysis

INSAT-3DR + IMDAA + GSMaP + DEM for continuous analysis of weather and terrain conditions.

2. Hyper-Local Multi-Hazard Prediction

Predicts thunderstorms, cloudbursts and flash floods with 0–100 risk scoring and multi-horizon forecasts.

3. Explainable Risk & Impact Mapping

Shows where the threat may move, why the risk is high and how terrain can increase flood impact.

4. Smart Verified Alerts

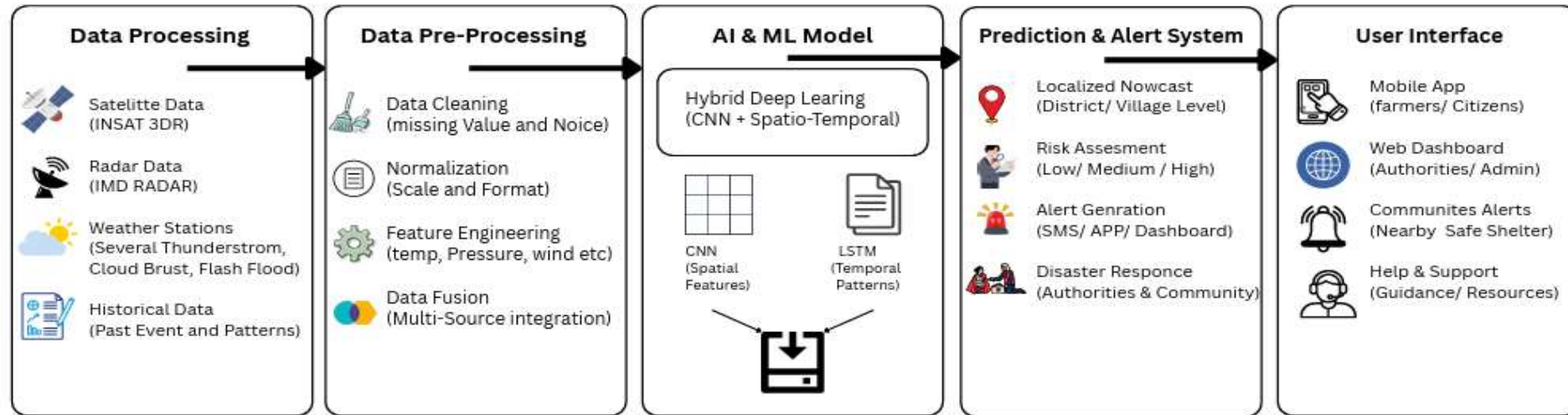
Admin-approved multilingual warnings with risk level, recommended action and nearby safe shelters.

DATA INPUTS

- ❑ **INSAT-3DR (ISRO) – Primary Satellite**
- ❑ **WV:** 6.5–7.0 μm → Moisture
- ❑ **TIR-1:** 10.2–11.2 μm → Cloud-Top Temperature
- ❑ **TIR-2:** 11.5–12.5 μm → Cloud/thermal features
- ❑ **MIR:** 3.8–4.0 μm → Cloud/thermal features
- ❑ **Other Data**
- ❑ **IMDAA:** Temperature, RH, U/V Wind → CAPE, CIN, Wind Shear
- ❑ **GSMaP-ISRO:** Rainfall
- ❑ **CartoDEM/SRTM:** Elevation & Drainage

Video
Link

SYSTEM ARCHITECTURE



DEMO WEB PAGE



TECHNICAL STACK

Frontend

Flutter (App Dev) HTML, CSS, JS

AI/ML

Tensor Flow (Deep learning) OPEN CV (Image Processing)

Backend

Python (Fast API) Node.js

Deployment

Render (Frontend deployment) Railways (backend Deployment)

Database/ Auth

Firebase (Realtime DB, Auth) SQLite (vector database)

OTHER

Github (Version Control) Docker (containerization)

IMPLEMENTATION PROCESS

Data Collection

- Satellite/radar
- Weather stations
- historical Data

preprocessing

- Cleaning and Normalization
- Handel missing Value
- Extract Feature

Model Development

- build CNN+ LSTM
- Train & Validate
- Tune Hyperparameters

Integration

- Connect API's
- Setup Database
- Deploy on Cloud

Testing & Deployment

- Validate Accuracy
- Run End-to-End tests
- Launch App + Dashboard

Challenges	Proposed Solutions
Different data sources and formats	Combine and align INSAT-3DR, IMDAA, GSMAp and DEM data.
Rapidly changing severe weather	Use AI-based multi-horizon nowcasting for early risk prediction.
Rare cloudburst and flash-flood events	Train the model using historical severe-weather events and handle class imbalance.
Multiple factors affect weather risk	Combine moisture, CAPE, CIN, wind shear, rainfall and cloud features.
Rainfall impact varies by terrain	Use DEM-based elevation, slope and drainage features for flood-risk mapping.
False or unnecessary alerts	Use confidence scoring and human/admin verification before alerts.

FEASIBILITY

- ☐ Uses available weather and satellite datasets.
- ☐ Developable using Python and open-source AI tools.
- ☐ Existing GIS tools can be used for risk mapping.
- ☐ Historical weather data can be used to train and test the model.
- ☐ The system can first be tested for one district and expanded later.

VIABILITY

- ☐ Helps authorities identify risky areas before severe weather.
- ☐ Reduces dependence on general district-level warnings.
- ☐ Can provide location-specific alerts to people.
- ☐ Can be integrated with existing disaster-management systems.
- ☐ Can be expanded to more districts and weather hazards.



BENEFITS OF SOLUTION



💰 ECONOMIC BENEFITS

Reduces Infrastructure Losses – Early action can protect roads, buildings and public assets.

Protects Local Businesses – Gives businesses time to secure assets and operations.

Reduces Emergency Response Costs – Targeted warnings can help authorities prioritize resources.

Protects Livelihoods – Supports timely action for workers, farmers and other weather-sensitive activities.

👤 SOCIETAL BENEFITS

Saves Lives – Timely, actionable warnings before severe weather impacts.

Hyper-Local Safety – Alerts specific high-risk micro-sectors instead of entire cities.

Inclusive Protection – Multilingual alerts (English, Marathi & Hindi) reach diverse communities.

Smarter Evacuation – AI-supported risk zones, safe shelters and evacuation routes.

🌿 ENVIRONMENTAL BENEFITS

Reduced Flood & Water Damage – Early warnings enable preventive action in high-risk zones.

Protection of Sensitive Areas – Helps identify vulnerable low-lying and environmentally sensitive locations.

Better Water Management – Rainfall and drainage-risk information supports smarter local water planning.

POTENTIAL IMPACT



Public Safety

- Reduce loss of life.
- Improve emergency preparedness.



Localized Protection

- Identify high-risk areas.
- Support targeted evacuation.



Faster Response

- Enable quicker alerts.
- Improve disaster coordination.

Sr. No.	Research Paper	Technique / Data Used	Key Finding	Relevance to AGNI-CAST
1	Leinonen et al. (2023) – Paper Link	Deep Learning, Satellite, NWP, DEM, multi-source fusion	Developed a multi-hazard thunderstorm nowcasting model using multiple weather and terrain data sources.	Supports multi-source fusion, multi-hazard prediction, DEM and hyper-local risk mapping.
2	Shukla et al. (2025) – Paper Link	INSAT-3D + ConvLSTM	Used ConvLSTM with INSAT-3D data for short-term rainfall nowcasting over the Bay of Bengal.	Supports INSAT + ConvLSTM-based nowcasting.
3	Priya et al. (2024) – Paper Link	INSAT-3D + temperature/RH profiles	INSAT-3D atmospheric profiles improved severe-thunderstorm prediction over eastern India.	Supports Indian context, INSAT data and thunderstorm prediction.
4	Piran et al. (2024) – Paper Link	Transformer + precipitation nowcasting	Proposed a Transformer-based model for precipitation nowcasting and disaster preparedness.	Supports your Spatial-Temporal Transformer approach.
5	Geoscience Frontiers (2021) – Paper Link	CNN + RNN + spatial flood mapping	Used deep learning to generate spatially explicit flash-flood probability maps.	Supports flash-flood risk mapping and hyper-local prediction.
6	Yang et al. (2024) – Paper Link	Explainable / Interpretable ML	Reviews XAI methods such as SHAP, LIME and Grad-CAM for weather and climate prediction.	Supports Explainable Risk Score and transparent AI decisions.